

Bill of Materials

Table 1

Component name	Quantity	Description	Approx. Cost (₹)
Raspberry Pi 4/Zero W	1	Any version of RPi	4,500
RPi camera	1	Camera module	300
HDMI/RCA cable	1	For video output	200
Wires	Depends	For connection	30
Total cost			5,030

Coding

It is assumed that you have Raspberry Pi (RPi) ready with Raspbian OS and Python IDE and have access to the RPi desktop either with VNC or HDMI based display. (Note. We are using Python3 Environment and the code is compatible with Python3 and above versions only.) So, we may start with the installation of modules and libraries for this project.

Open the Linux terminal on your Raspberry Pi and install the required libraries using following commands:

```
sudo apt-get update
sudo apt-get upgrade
sudo nano /etc/dphys-swapfile
```

Change the line CONF_SWAPSIZE=100
to CONF_SWAPSIZE=1024

Continue with the following commands:

```
sudo /etc/init.d/dphys-swapfile stop
sudo /etc/init.d/dphys-swapfile start

sudo pip3 install opencv
sudo pip3 install numpy
wget https://bootstrap.pypa.io/get-pip.py
pip3 install dlib
pip3 install tensorflow
```

```
import os
import cv2
import datetime
import numpy as np
from picamera.array import PiRGBArray
from picamera import PiCamera
import tensorflow as tf
import argparse
import sys
import time

IM_WIDTH = 1280
IM_HEIGHT = 720
#IM_WIDTH = 640      Use smaller resolution for
#IM_HEIGHT = 480     slightly faster framerate
time=time.time()
# Select camera type (if user enters --usbcam when calling this script,
# a USB webcam will be used)
camera_type = 'picamera'
parser = argparse.ArgumentParser()
parser.add_argument('--usbcam', help='Use a USB webcam instead of picamera',
                    action='store_true')
args = parser.parse_args()
if args.usbcam:
    camera_type = 'usb'
```

Fig. 4: Code with modules and libraries

```
# Path to frozen detection graph .pb file, which contains the model that is used
# for object detection.
PATH_TO_CKPT = os.path.join(CMD_PATH, MODEL_NAME, 'frozen_inference_graph.pb')

# Path to label map file
PATH_TO_LABELS = os.path.join(CMD_PATH, 'data', 'New.pbtxt')

# Number of classes the object detector can identify
NUM_CLASSES = 90

## Load the label map.
# Label maps map indices to category names, so that when the convolution
# network predicts '5', we know that this corresponds to 'airplane'.
# Here we use internal utility functions, but anything that returns a
# dictionary mapping integers to appropriate string labels would be fine
label_map = label_map_util.load_labelmap(PATH_TO_LABELS)
categories = label_map_util.convert_label_map_to_categories(label_map, max_num_classes=NUM_CLASSES, use_display_name=True)
category_index = label_map_util.create_category_index(categories)
```

Fig. 5: Defining path for the new label file

```
cv2.putText(frame, "FPS: {0:.2f}".format(frame_rate_calc), (30, 50), font, 1, (255, 255, 0), 2, cv2.LINE_AA)
name= ([category_index.get(value) for index,value in enumerate(classes[0]) if scores[0,index] > 0.5])
print (name)
data=str(name)
string=([[]])
string2=([None])
imagestring=str(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))

#Print (name)
if (name != string1):
    print("Animal detected")
    text_files=open("frozen_db.txt","a+")
    n= text_file.write(data)
    n= text_file.write(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))
    n= text_file.write("-----")
    img_name=(imagestring + ".jpg")
    path="/home/pi/tensorflow/models/research/object_detection/databasefolder"
    cv2.imwrite(img_name,frame)
    cv2.imwrite("/home/pi/tensorflow/models/research/object_detection/databasefolder/"+img_name,frame)
```

Fig. 6: Saving the detected image with timestamp in a defined path folder

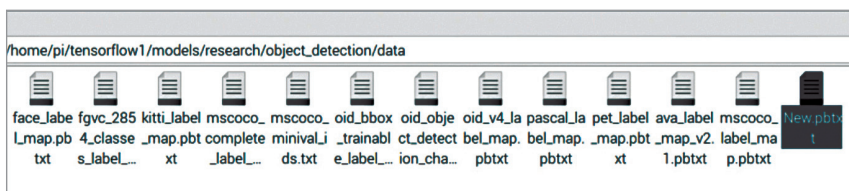


Fig. 7: Creating file named New.pbtxt under data folder

```
File Edit Format Run Options Window Help

item {
  name: "/m/015p6"
  id: 16
  display_name: "bird"
}
item {
  name: "/m/01yrx"
  id: 17
  display_name: "cat"
}
item {
  name: "/m/0bt9lr"
  id: 18
  display_name: "dog"
}
item {
  name: "/m/03k3r"
  id: 19
  display_name: "horse"
}
item {
  name: "/m/07b9p"
  id: 20
  display_name: "sheep"
}
item {
  name: "/m/01xq0k1"
  id: 21
  display_name: "cow"
}
item {
  name: "/m/0bwd_0j"
  id: 22
  display_name: "elephant"
}
item {
  name: "/m/01dws"
  id: 23
  display_name: "bear"
}
item {
  name: "/m/0898b"
  id: 24
  display_name: "zebra"
}
```

Fig. 8: Creating category of animals/species for display in New.pbtxt